

BVER: BIDIRECTIONAL INCREMENTAL VORONOI-BIASED EXPLORATION CURRICULUM FOR REINFORCEMENT LEARNING

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Paper under double-blind review

ABSTRACT

Long-horizon tasks with sparse rewards pose an exploration bottleneck for goal-conditioned reinforcement learning: a policy started from the initial state almost never reaches the goal and receives no learning signal. In locomotion and manipulation, this is usually sidestepped with reference motions, hand-designed curricula, or shaped rewards, each requiring demonstrations or task-specific engineering. Automatic start-state or goal curricula avoid this design burden by generating intermediate tasks, but existing methods typically expand from one side only, proposing goals for a fixed start or start states for a fixed goal, so the full distance to the target must be covered from that single direction. We propose BVER¹, an automatic curriculum that instead expands from both ends at once. Inspired by bidirectional RRT planning, BVER grows a set of start states outward from the goal and a set of goals outward from the initial state distribution, biases each toward unexplored task space, and steers the two toward each other; one goal-conditioned policy is trained on tasks drawn from both sets. On sparse-reward quadrupedal box climbing, a task hard enough that no solution emerges without a curriculum, BVER reaches 95% success in roughly 65% fewer training iterations than the best demonstration-free baseline, and the resulting policy holds up under wide start, goal, and yaw variation. Ablations confirm that expanding from both ends outperforms either direction alone, and the same curriculum solves sparse-reward point-mass mazes.

1 INTRODUCTION

Many robotic tasks are posed as reaching a goal: a quadruped that should climb onto a box is rewarded for standing on top of it, not for how it gets there. Under such a sparse reward and a long horizon, a policy started from the initial state distribution essentially never reaches the goal and therefore receives no learning signal (Florensa et al., 2017). Policy gradient methods such as PPO (Schulman et al., 2017) perform a local search that improves the policy from the returns of the trajectories it currently generates, and a reward that none of these trajectories collects cannot guide it.

In quadrupedal locomotion this bottleneck is usually avoided rather than solved. Terrain curricula with dense rewards (Rudin et al., 2022; Lee et al., 2020), reference motions from trajectory optimization (Fuchioka et al., 2023), and hand-designed curricula (Atanassov et al., 2024) supply the missing signal, but each requires a demonstration or engineering specific to one task, and curiosity-driven exploration (Schwarke et al., 2023) adds rewards that the task itself does not provide.

Automatic curriculum learning replaces this design effort with intermediate tasks generated during training (Narvekar et al., 2020; Portelas et al., 2020b). Start-state and goal curricula address goal-conditioned tasks directly by shortening the task from one of its two ends: start-state curricula begin closer to the goal, with start states taken from a reference trajectory (Peng et al., 2018; Resnick et al., 2018) or from random walks outward from the goal (Florensa et al., 2017), and goal curricula keep the start fixed and propose intermediate goals (Florensa et al., 2018; Ren et al., 2019; Cho et al.,

¹Project website, videos and code: https://juripfammatter.com/html/bver_demo/bver.html

2023). Both typically expand from one end only, so the curriculum has to carry its frontier across the entire distance to the end that stays fixed. The policy, in turn, sees the target task from one side, reaching the goal from many starts or many goals from the start, but not both. In addition, the random walks of reverse curriculum generation (RC; Florensa et al., 2017) are seeded without regard to coverage, so nothing promotes the frontier into unvisited task space or toward the initial state distribution.

Two mechanisms from sampling-based motion planning address exactly this. Rapidly-exploring random trees (RRT; LaValle & Kuffner Jr, 2001) extend the vertex nearest to a uniform sample of the space, so growth is biased toward large unexplored regions (the Voronoi bias), and RRT-Connect (Kuffner & LaValle, 2000) grows one tree from the start and one from the goal and extends each toward the other until they meet. BVER (Bidirectional incremental Voronoi-biased Exploration curriculum for Reinforcement learning) transfers both to curriculum generation. It grows start states outward from the goal by random walks (the backwards expansion) and goals outward from the initial state distribution in the same way (the forwards expansion), connects the two sets, and trains one goal-conditioned policy on both intermediate starts and goals. This creates a curriculum whose proposed tasks are both at the right difficulty and promote coverage of the task space, and thus of the target task.

We evaluate BVER on simulated quadrupedal box climbing with sparse rewards, where PPO finds no solution without a curriculum. BVER converges faster than every reference-free curriculum we compare against, and the resulting policy holds up under start, goal, and yaw variations on which the other curricula fail. Ablations show that expanding from both ends outperforms either direction alone, and the same curriculum solves four scanned real-world terrains with unchanged curriculum hyperparameters, as well as three point-mass mazes.

Our contributions are:

- An expansion mechanism carrying the Voronoi bias of RRT and the connect step of RRT-Connect over to curriculum generation, growing each frontier toward unexplored task space and toward the other frontier.
- An automatic start-goal curriculum for goal-conditioned reinforcement learning with sparse rewards and long horizons that expands intermediate start states and intermediate goals at the same time and trains one policy on both, without reference motions, shaped rewards, or a hand-designed curriculum.
- An evaluation against reference-free and reference-based curricula on high-dimensional quadrupedal locomotion with sparse rewards and long horizons, ablations of both expansion directions and of the connect step, and a study of route choice on a terrain with three routes.

In total, we evaluate on twelve tasks: four box climbing variants, the three-route terrain, four scanned real-world terrains, and three point-mass mazes.

2 RELATED WORK

Automatic curriculum learning: Automatic curriculum learning (ACL) adapts the sequence of training tasks to the capabilities of the learner during training (Narvekar et al., 2020; Portelas et al., 2020b). Portelas et al. (2020b) sort ACL methods for data collection by what they control: the initial state, the reward function, the goal, the environment, or the opponent. Environment curricula sample the parameters of a procedural generator by absolute learning progress (Portelas et al., 2020a), replay previously seen levels prioritized by the magnitude of their value loss (Jiang et al., 2021), or widen the ranges of domain randomization once performance at their boundary exceeds a threshold (Akkaya et al., 2019); all of them assume an environment that can be parameterized or regenerated. Reward curricula such as RND (Burda et al., 2018) add as intrinsic reward the error of a network that predicts the features of a fixed random network, which by its authors’ own account handles local exploration but not global exploration over long horizons. BVER controls neither environment nor reward and changes only where an episode starts and which goal it is given.

Start-state and goal curricula: Reverse curriculum generation (RC; Florensa et al., 2017) grows start states outward from a given goal state with short rollouts of random actions and keeps those from which the current policy reaches the goal sometimes but not always; it needs no demonstrations, only a simulator that resets to arbitrary states and one goal state. BaRC (Ivanovic et al., 2019) replaces the random rollouts with backward reachable sets computed from an approximate dynamics model, so that the frontier grows in every direction the dynamics allow. Reference state initialization (RSI; Peng et al., 2018) samples the start state uniformly from a reference motion, Backplay (Resnick et al., 2018) moves a window of start states backward along a single demonstration on a fixed schedule, and RFCL (Tao et al., 2024) runs a reverse curriculum per demonstration and then prioritizes initial states of the task by success rate; all three vary only the start state and rely on demonstrations. Goal curricula generate goals for the task’s own initial state distribution: GoalGAN (Florensa et al., 2018) trains a generator to propose goals of intermediate difficulty, HGG (Ren et al., 2019) selects achieved states from the replay buffer by their Wasserstein distance to the target task distribution and their value, and OUTPACE (Cho et al., 2023) selects them by temporal distance from the initial distribution, uncertainty, and progress toward given outcome examples. In contrast, BVER grows intermediate starts and intermediate goals at the same time, biases both toward unexplored task space, and connects the two sets, which, unlike RC and BaRC, drives the frontiers toward the target task. Combining start-state generation with goal generation, by analogy to RRT-Connect, is what Florensa et al. (2017) proposed as future work.

Sampling-based planning: A rapidly-exploring random tree (LaValle & Kuffner Jr, 2001) extends the vertex nearest to a random sample of the space; since a vertex is selected with probability proportional to the area of its Voronoi region, the tree is biased to explore rapidly (Kuffner & LaValle, 2000). RRT-Connect (Kuffner & LaValle, 2000) grows two trees from the start and the goal and tries to connect them. BVER seeds its random walks with the same Voronoi bias and connect step, growing a curriculum rather than a path.

Quadrupedal locomotion: Current state-of-the-art quadrupedal locomotion methods do not train on sparse-reward, long-horizon tasks and often rely on reference motions or hand-designed curricula. For example, Rudin et al. (2022) utilize a game-inspired terrain curriculum with dense rewards. Lee et al. (2020) replace the game-inspired curriculum with an adaptive, particle-filter-based parametric terrain curriculum. Fuchioka et al. (2023) use trajectory optimization to find a solution and then train with RSI (Peng et al., 2018) as an initialization curriculum. Atanassov et al. (2024) hand-design a three-stage curriculum for jumping and initialize the first stage at random heights and upward velocities, a form of RSI without a reference trajectory, which they find necessary to learn jumping in place. Schwarke et al. (2023) learn a quadrupedal loco-manipulation task on sparse rewards by extending RND (Burda et al., 2018) with a curiosity state. Our method differs from these as it learns on sparse rewards directly, without reference motions, hand-designed curricula, or additional rewards.

3 PRELIMINARIES

3.1 GOAL-CONDITIONED MARKOV DECISION PROCESS

We model the task as a goal-augmented Markov decision process (GA-MDP) (Liu et al., 2022) $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, P, r, \gamma, \rho_0, \mathcal{G}, \phi \rangle$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P(s_{t+1} | s_t, a_t)$, discount factor γ , initial state distribution ρ_0 , and goal space $\mathcal{G}$. The known mapping $\phi : \mathcal{S} \rightarrow \mathcal{G}$ projects a state onto the goal it achieves (Andrychowicz et al., 2017), and a goal $g \in \mathcal{G}$ is reached in state s if $\|\phi(s) - g\| \leq \epsilon$ for a tolerance ϵ . The reward is sparse and binary, $r(s_t, a_t, g) = \mathbf{1}[\|\phi(s_{t+1}) - g\| \leq \epsilon]$, episodes have a fixed horizon H , and reaching the goal does not terminate the episode, so the reward is collected in every step in which the goal condition holds.

The agent acts with a goal-conditioned policy $\pi_\theta(a | s, g)$, realized in the concatenated form of universal value function approximators (Schaul et al., 2015): a single network takes the concatenated vector $[s, g]$ as input, and so does the critic $V_\psi(s, g)$. Both are trained with PPO (Schulman et al., 2017) in massively parallel simulation (Rudin et al., 2022).

For a trajectory $\tau = (s_0, a_0, \dots, s_H)$ collected under goal g , we denote the discounted return by $R(\tau, g) = \sum_{t=0}^{H-1} \gamma^t r(s_t, a_t, g)$. Its expectation from a given start state is the universal value func-

tion (Schaul et al., 2015; Ren et al., 2019)

$$V^\pi(s_0, g) = \mathbb{E}_{a_t \sim \pi(\cdot | s_t, g), s_{t+1} \sim P(\cdot | s_t, a_t)} [R(\tau, g)]. \quad (1)$$

Because the reward keeps accruing at the goal, $V^\pi(s_0, g)$ measures how quickly and how reliably π reaches g from s_0 and stays there, rather than a binary success. Following Ren et al. (2019), a distribution ρ over start-goal pairs $(s_0, g) \in \mathcal{S} \times \mathcal{G}$ defines a *task*, and the performance of π on it is

$$J_\rho(\pi) = \mathbb{E}_{(s_0, g) \sim \rho} [V^\pi(s_0, g)]. \quad (2)$$

The *target task* consists of the initial state distribution ρ_0 and a single target goal $g^* \in \mathcal{G}$, i.e., $\rho^*(s_0, g) = \rho_0(s_0) \delta_{g^*}(g)$, and the learning objective is $\max_\theta J_{\rho^*}(\pi_\theta)$.

3.2 CURRICULUM

Optimizing J_{ρ^*} directly fails under sparse rewards and long horizons: a randomly initialized policy essentially never reaches g^* from ρ_0 and thus receives no learning signal (Florensa et al., 2017). Curriculum learning instead trains on a sequence of easier tasks (Narvekar et al., 2020; Portelas et al., 2020b). Following Tao et al. (2024), a *curriculum* is a sequence of task distributions over start-goal pairs,

$$\rho_1, \rho_2, \dots, \rho_E, \quad \rho_e \in \Delta(\mathcal{S} \times \mathcal{G}), \quad (3)$$

where ρ_e is the distribution from which the start-goal pairs of training iteration e are drawn, together with an *advancement criterion* that constructs ρ_{e+1} from ρ_e and the experience collected with the current policy π_e . The intermediate tasks are auxiliary: the objective remains that of Section 3.1, and the curriculum is chosen such that the policy obtained after E iterations maximizes $J_{\rho^*}(\pi_E)$ (Portelas et al., 2020b).

Besides a start-state curriculum $\rho_e(s_0, g) = \rho_e^S(s_0) \delta_{g^*}(g)$ (Florensa et al., 2017; Tao et al., 2024) or a goal curriculum $\rho_e(s_0, g) = \rho_0(s_0) \rho_e^G(g)$ (Florensa et al., 2018; Ren et al., 2019), the joint formulation in Eq. (3) also admits the mixture

$$\rho_e(s_0, g) = \beta \rho_e^S(s_0) \delta_{g^*}(g) + (1 - \beta) \rho_0(s_0) \rho_e^G(g), \quad \beta \in [0, 1], \quad (4)$$

which trains one shared goal-conditioned policy on intermediate starts and intermediate goals at the same time; $\beta = 1$ and $\beta = 0$ recover the unidirectional cases, and $\beta = \frac{1}{2}$ is the form our method instantiates (Section 4).

3.3 ASSUMPTIONS

Analogous to Florensa et al. (2017), we assume that (A1) the simulator can be reset to arbitrary states $s \in \mathcal{S}$; (A2) random-walk rollouts can be simulated from any state, i.e., we can sample from P under random actions; (A3) the initial distribution ρ_0 can be sampled and one goal state $s^g \in \mathcal{S}$ with $\phi(s^g) = g^*$ is known; and (A4) a bounded task space $\mathcal{T} \subseteq \mathcal{G}$ that contains the target task is known and can be sampled uniformly.

4 METHOD

Our method maintains finite sets of states from which the starts and goals of each training iteration are sampled uniformly at fixed ratios; ρ_e^S and ρ_e^G in Eq. (4) are the resulting discrete distributions, and we describe the method in terms of these sets. In the following, we first describe our method’s frontier expansion mechanism for the backwards expansion, and how it extends RC’s expansion mechanism to provide coverage of the task space and thus the initial distribution. Then we show how the same mechanism can be applied to the forwards expansion by substituting the start state with the goal state. Finally, we combine the two into our main contribution: a bidirectional expansion mechanism that expands the curriculum from both the start and goal states, leading to more sample-efficient and robust learning. An overview of the full method is given in Figure 9 in the appendix.

4.1 INCREMENTAL SPATIAL EXPANSION

Exploration: Similar to RC (Florensa et al., 2017), we initialize the policy training with valid states obtained from simulating random actions in state space (Figure 1a). After the training iteration, we separate frontier states ($R_{\min} < R < R_{\max}$) and successful states ($R_{\max} \leq R$), where

$R = R(\tau, g)$ is the return of a single rollout, i.e., a one-sample estimate of $V^\pi(s_0, g)$, and then mix them with fixed ratios to create a balanced training set. This set contains mostly states that are neither too hard nor too easy for the current policy, and we call this set the “mixed” training set.

When choosing seeds for the next iteration of random walks, we add an additional criterion to enforce coverage of the task space and thus promote convergence to the initial distribution. To do so, we find the spatial frontier (Figure 1b) of the explored state space by (1) sampling a set of random points uniformly from the task space $\mathcal{T}$, (2) computing the nearest neighbor in the mixed training set for each of the sampled states, and (3) selecting these states as the seeds for the random walks (Figure 1c). Similarly to the Voronoi bias inherent in Rapidly-exploring Random Trees (RRT) (LaValle & Kuffner Jr, 2001), this biases the random walks towards unexplored areas of the task space, as they have a higher probability of being sampled than states in already explored areas.

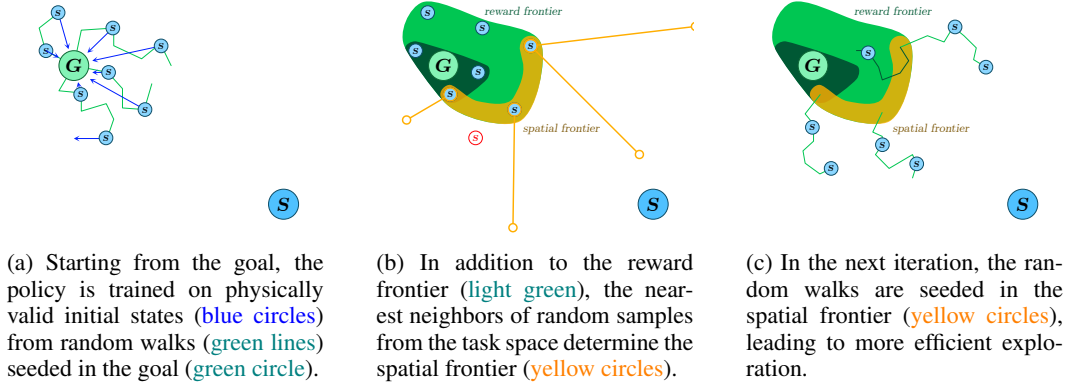


Figure 1: Backwards expansion mechanism used in the exploration step.

Exploitation: In addition to training the policy on unseen states in the exploration step, we also continue training on the mixed training set. This has two effects: (1) training on the reward frontier improves the policy, and (2) training on successful states mitigates catastrophic forgetting. A fixed ratio of training iterations is allocated to this step. If the policy is trained on a parallelized environment, this can be done in parallel with the exploration step, which is the case in our experiments.

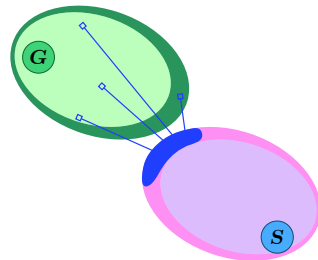
Both the exploration and exploitation steps are repeated until the policy is able to solve the task from the initial distribution.

4.2 BIDIRECTIONAL EXPANSION

Forwards expansion: The backwards expansion mechanism described in the previous section can be applied to the forwards expansion by substituting the start state with the goal state. The goal-conditioned policy is trained on initial states sampled from the initial distribution ρ_0 and goals sampled from the spatial frontier of the explored state space, i.e., $(s_0 \sim \rho_0, g_i)$ rather than (s_i, g^*) . The exploration and exploitation steps are then identical, with the minor difference that the random walks are seeded in states that can be reached rather than states from which the goal can be reached.

Bidirectional expansion: Combining the two expansion mechanisms, we train one shared goal-conditioned policy on both $(s_0 \sim \rho_0, g_i)$ and (s_i, g^*) , with the training budget split evenly between the two expansions ($\beta = \frac{1}{2}$ in Eq. (4)). This leads to a diverse, incrementally expanding curriculum that covers the reachable task space from two directions. As a result, the policy is trained to reach the target goal from a wide range of start states, and to reach a wide range of goals from the initial distribution, leading to more robust policies.

Connecting the two frontiers: In addition to the bidirectional expansion, we can also connect the two frontiers by replacing the random samples from $\mathcal{T}$ in step (1) of the spatial expansion with samples from the other frontier (Figure 2). For example, when expanding the forwards frontier, we sample states from the mixed training



set of the backwards expansion and then find their nearest neighbors in the mixed training set of the forwards expansion. As a result, the seeds of the forwards random walks lie on the side of the forwards set that faces the backwards set, and the intermediate goals grow toward the target goal g^* . Symmetrically, the seeds of the backwards random walks face the forwards set, and the intermediate starts grow toward the initial distribution ρ_0 .

This leads to a more sample-efficient exploration, as each expansion moves toward the region the other expansion has already made solvable instead of spreading uniformly over $\mathcal{T}$. This is similar to the RRT-Connect algorithm (Kuffner & LaValle, 2000), where two trees are grown towards each other until they connect.

To promote both even coverage of the task space and the connection of the two frontiers, we sample from both the other frontier and the task space with a fixed ratio, the connect ratio. It balances even coverage of $\mathcal{T}$ against the connection of the two frontiers and is a hyperparameter of our method.

5 EXPERIMENTS

We conduct four experiments to answer the following research questions:

1. Does our method outperform state-of-the-art curriculum learning methods on a high-dimensional quadrupedal locomotion task with sparse rewards and long horizons?
2. Does expanding the curriculum bidirectionally improve sample efficiency and robustness compared to unidirectional expansion?
3. Does task-space coverage-driven exploration promote more diverse learning outcomes?
4. Does our method transfer to real-world terrains and tasks, demonstrating its practical applicability and robustness in real-world scenarios?

The experiments are implemented in Isaac Lab (NVIDIA et al., 2025) with the ANYmal D quadrupedal agent and the RSL-RL framework (Schwarke et al., 2025) for training the policy with PPO (Schulman et al., 2017). The following experiments use standard observation and action spaces which are detailed in Appendix A.1. The tasks are shown in Figure 3. The task space $\mathcal{T}$ used for our method’s random point sampling and goal definition is a 3D box defined in the (x, y, z) space, i.e., $\phi(s)$ is the projection of the base position to (x, y, z) . We found this to be the most effective task space, as it is both convenient for goal definition and provides a simple Euclidean distance metric for the nearest-neighbor search.

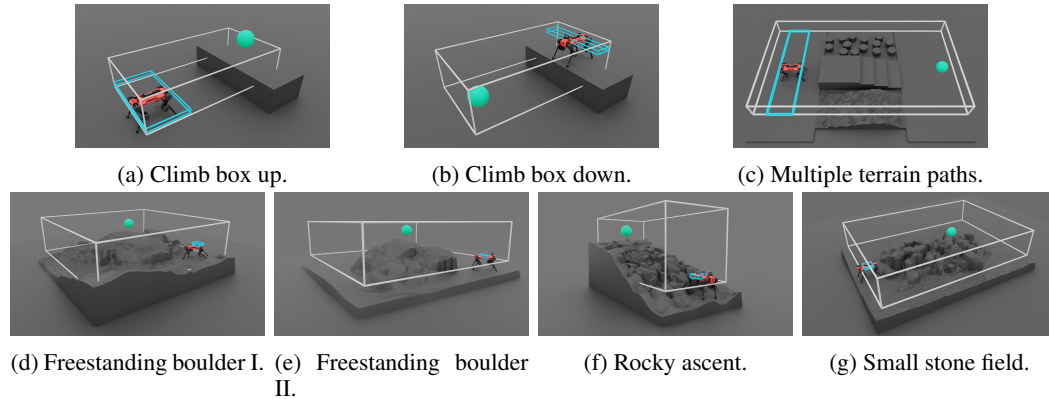


Figure 3: Experimental setup. Cyan indicates the initial distribution, green the target goal, and the black/white box represents the task space.

In all experiments, the rewards are defined as a combination of the sparse goal reward from Section 3.1 with $\epsilon = 0.25$ m, i.e., the agent has to reach the goal within a certain distance, and a termination penalty for collisions of the base with the terrain.

In the *multiple terrain paths* experiments, a potential-based reward shaping term (Ng et al., 1999), which leaves the set of optimal policies of the underlying GA-MDP unchanged, is added to provide a reward signal strong enough for the no-curriculum PPO baseline to find solutions:

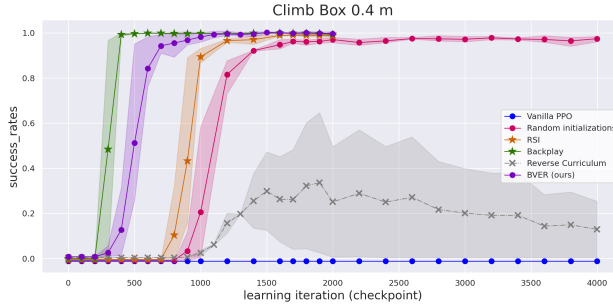
$$r_{\text{progress}}(s_t, s_{t+1}, g) = \|\phi(s_t) - g\| - \gamma \|\phi(s_{t+1}) - g\|$$

5.1 SAMPLE EFFICIENCY AND ROBUSTNESS

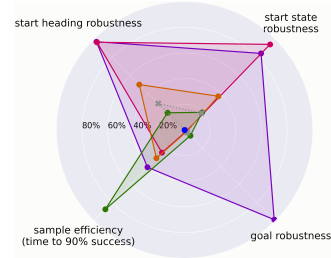
To evaluate sample efficiency and robustness, we compare our method BVER against state-of-the-art curriculum learning methods on a sparse-reward box climbing task. The task is defined as a goal-conditioned locomotion task where the agent has to reach a target goal position on top of a box from an initial distribution of start positions in front of the box. The task is challenging due to the sparse reward signal and the long horizon required to reach the goal.

Sample Efficiency: The results in Figure 4a already show a clear distinction between the vanilla PPO baseline, the curricula and the imitation learning methods (marked with *). The policies trained on the initial distributions and no curriculum (vanilla PPO) are unable to solve the task, while training with random starts achieves a success rate of more than 95%. This shows that the reward signal is attainable and the task solvable by initializing the agent closer to the goal. Backplay (Resnick et al., 2018) shows the fastest convergence, and its advantage over RSI (Peng et al., 2018) shows the importance of ordering the task over training time. For the reference-free curricula, our method BVER shows the fastest convergence and highest success rate, outperforming both RC (Florensa et al., 2017) and random starts, while even surpassing RSI. BVER reaches a success rate of 95% in roughly 65% fewer training iterations than the best reference-free baseline.

Robustness: We evaluate the trained policies on (1) perturbations to the yaw angle used in the initialization, (2) variations in start positions (goal set to the target goal), and (3) variations in goal positions (initialized on the initial distribution). The results in Figure 4b show that BVER’s bi-directional curriculum leads to policies that can solve the task under various start-goal configurations, while the other curricula fail to generalize to unseen configurations.



(a) BVER is able to achieve a success rate of over 95% faster than all other reference-free curricula and even faster than one reference-based method (RSI).



(b) BVER’s start-goal curriculum leads to policies that can solve the task under various start-goal configurations.

Figure 4: Sample efficiency and robustness evaluated on a sparse-reward box climbing task.

5.2 ABLATIONS: FORWARDS / BACKWARDS SYMMETRY

Maze ablations: To evaluate the contribution of individual components of BVER, we conduct further ablation studies on three maze environments (Figure 5a) similar to those used in prior work (Florensa et al., 2017; Cho et al., 2023). Specifically, the agent (orange point mass) has to find a path from the initial distribution (cyan rectangle) to the goal (green ball). First, we evaluate the effect

of our curriculum against a baseline that does not use any curriculum (*vanilla PPO*) and a simple curriculum that samples intermediate goals (to reach from the start) and intermediate starts (from which to reach the target goal) randomly from the free space (*random initializations*). The results in Figure 5b show that both curricula enable finding solutions in all three environments, whereas vanilla PPO does not. In the more complex *N-maze* and *large-maze* environments, BVER shows faster convergence than the random curriculum.

We compare using only the forwards expansion, only the backwards expansion, and the bidirectional expansion in Figure 5c. In addition, we evaluate the effect of connecting the two frontiers by comparing a version of BVER that only samples from the task space to a version that samples from both the task space and the other frontier. The results in Figure 5d show that connecting the two frontiers leads to better sample efficiency in the cases where the optimal path covers only a small fraction of the total search space (*N-maze* and *large-maze*). However, there is no significant difference between connect ratios above 0.5.

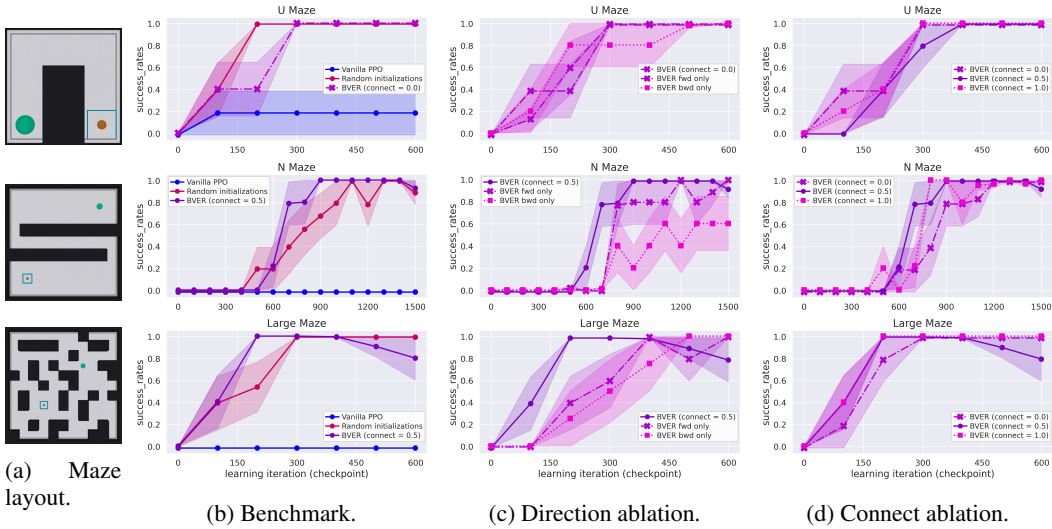


Figure 5: Comparison of BVER against vanilla PPO and a random curriculum, and ablations of different components (expansion directions) and hyperparameters (connect ratio).

Box climbing ablations: We evaluate the effect of the bidirectional expansion mechanism by comparing BVER on two symmetric climbing tasks (up and down) against its unidirectional variants, i.e., only expanding the curriculum from the start state (forwards expansion) or only expanding the curriculum from the goal state (backwards expansion). The results in Figure 6 show that combining forwards and backwards expansions leads to the best results irrespective of the task (up or down) and box height. In both tasks, the expansion that starts on top of the box performs better than the one that starts on the ground. This is likely because the random walks are subject to gravity, so traversing down the box edge is more likely than traversing up it.

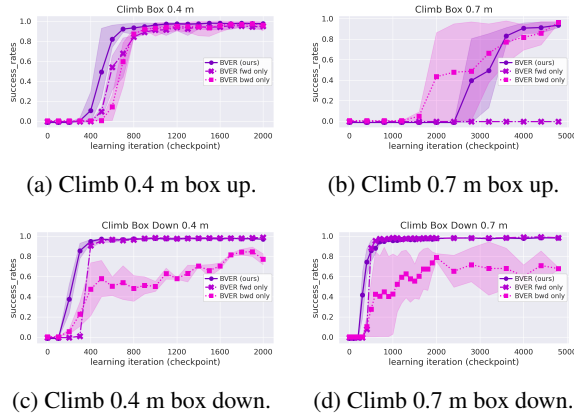


Figure 6: Ablations on the box climbing task show that combining forwards and backwards expansions leads to the best results irrespective of direction (up or down) and box height.

5.3 EMERGENT BEHAVIOR: PATH DIVERSITY

To answer research question 3, we design a task with three qualitatively different routes from the initial distribution to the goal: a set of stairs in the center, a field of pillars on one side, and a ramp on the other (Figure 3c). The initial distribution is chosen to span the full width of the terrain, such that each of the three routes is the closest one for roughly a third of the initial states and no route is favored. After training, we roll out both policies from this broad initial distribution and classify every successful episode by the route it takes.

Figure 7 shows that PPO without a curriculum collapses onto a single behavior: all successful episodes traverse the ramp, which is the easiest but not necessarily the shortest route, regardless of where the episode starts. Once this route is discovered, exploration of the remaining task space stops, and the policy never learns to climb the stairs or navigate the pillars, even when it is initialized directly in front of them. In contrast, BVER’s task-space coverage-driven expansion forces the curriculum to remain active in all regions of the task space, and the resulting policy uses all three routes, choosing the one closest to its initial state. This indicates that task-space coverage promotes more diverse learning outcomes: rather than committing to the first solution it finds, the policy acquires the full set of locomotion skills the terrain affords, which makes it less dependent on any single route being available.

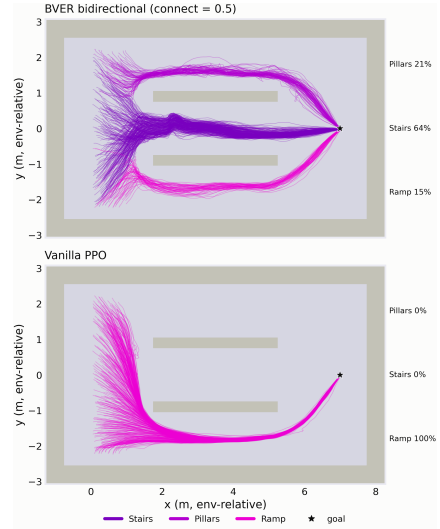


Figure 7: Path diversity on the three-route terrain. PPO without a curriculum collapses onto the ramp, while BVER learns to traverse all three routes.

5.4 REAL-WORLD TERRAINS

To show its practical applicability and answer research question 4, we evaluate BVER on four different real-world terrains with the ANYmal D quadrupedal robot in an identical setup to the box climbing task in Section 5.1. We use the same hyperparameters and curriculum settings as in the box climbing task, and choose the task space to roughly cover the terrain area. The four terrains in Figure 3 can be categorized as follows: (1) climbing a freestanding boulder (Figures 3d and 3e), (2) traversing over a field of small stones (Figure 3g), and (3) ascending a steep rocky slope (Figure 3f), each requiring different locomotion skills. On the *freestanding boulders* and the *small stone field*, the initial distribution is chosen such that no specific path to the goal is favored and the policy has to find its own path to the goal. Table 1 shows that, similar to box climbing, PPO without a curriculum is unable to solve the task, while BVER finds solutions in reasonable time on all four terrains.

Table 1: Training iterations needed to reach a success rate of 95% on the four real-world terrains.

Task	Freestanding boulder I	Freestanding boulder II	Small stone field	Rocky ascent
Vanilla PPO	—	—	—	—
BVER	$1800 \pm x$	$2300 \pm x$	$2500 \pm x$	$3500 \pm x$

Figure 8 shows the explored regions on the *rocky ascent* terrain at different training iterations. The intermediate starts and goals are clustered around their respective roots at the beginning of training, and gradually expand to cover the entire task space as the policy’s skill improves. Even though the task space (black/white box) is much larger than the reachable area of the robot, the curriculum is able to find a path to the goal without any prior knowledge of the terrain, showing the effectiveness of the random walks. The even spread of intermediate starts and goals across the task space indicates that the policy has learned to traverse the terrain rather than just solving the target task.

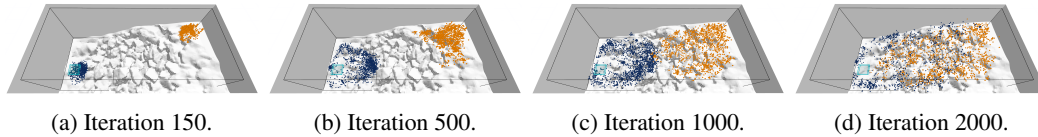


Figure 8: Samples of the explored regions on the *rocky ascent* terrain at different training iterations. **Blue** indicates the intermediate goals (forwards expansion), **orange** indicates the intermediate starts (backwards expansion), **cyan** indicates the initial distribution, and the black/white box represents the task space.

5.5 GENERALIZATION TO DIFFERENT DOMAINS: MANIPULATION TASKS

6 CONCLUSION

We propose a method that combines both start-state and goal curricula for sparse-reward, long-horizon goal-conditioned reinforcement learning. We transfer the idea of bidirectional expansion from classic sampling-based planning (LaValle & Kuffner Jr, 2001; Kuffner & LaValle, 2000) to curriculum generation, growing a set of start states outward from the goal and a set of goals outward from the initial state distribution, biasing both toward unexplored regions of the task space, and steering the two frontiers toward each other. Experimental results show that this leads to sample-efficient and robust policies on quadrupedal box climbing, real-world terrains, point-mass mazes, and manipulation tasks. Furthermore, our approach adds minimal computational overhead and thus does not require excessive training time.

Our method relies on a simulator that can be reset to arbitrary states, on rollouts of random actions, and on a bounded task space with a meaningful distance metric. For the latter, we rely on a Euclidean distance metric in this work, but future work could extend to or even learn different distance metrics, for example the observed number of time steps between two states proposed by Hartikainen et al. (2020). Instead of using random walk rollouts to estimate the reachability of states, future work could also explore the use of learned dynamics models or using the current policy to roll out trajectories.

AI USE STATEMENT

REPRODUCIBILITY STATEMENT

Experiments can be reproduced with the code provided in the project website. The code includes all hyperparameters, random seeds, and instructions to run the experiments. We run all experiments on a single NVIDIA RTX 4090 GPU with six different random seeds each.

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A APPENDIX

A.1 SIMULATION SETUP

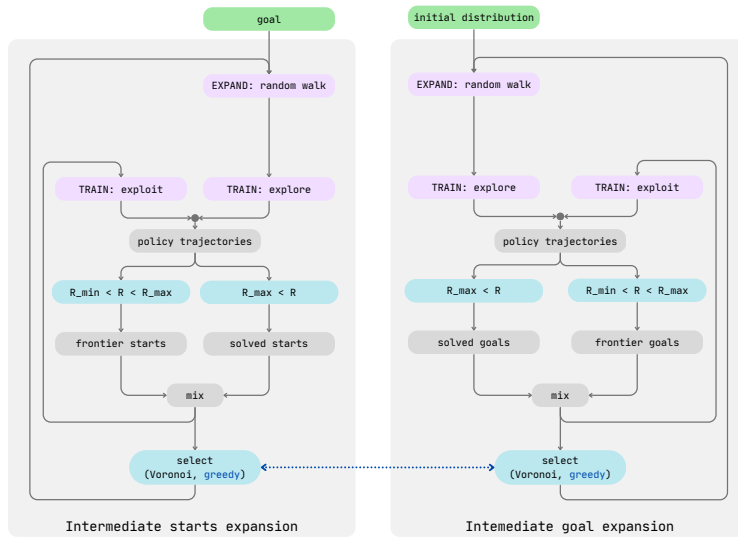


Figure 9: Overview of BVER.